HomeScope

AI-Powered American House Price Dashboard

A one-month Project & Portfolio IV pitch using real estate datasets, market averages, data visualization, and machine learning price estimation.

User Value

Check average prices by city, zip code, bedrooms, bathrooms, and square footage.

AI Value

Predict a fair price range and flag listings that appear above or below market.

Problem

Home prices are hard to understand without context.

Price Alone Is Misleading

A $350,000 listing may be affordable in one city and overpriced in another. Users need local comparisons, not just national averages.

Market Data Is Scattered

Listings, demographics, trends, and national price history are usually spread across multiple websites and dashboards.

Users Need Simple Signals

Most people want to know: Is this home close to the average? Is it above market? What factors are pushing the price?

Project goal:

Build a focused dashboard that turns housing data into average price insights, visual comparisons, and a prediction model that can be evaluated with real metrics.

Solution Concept

A dashboard built around three user pillars.

1. Explore Market Averages

Search by state, city, zip code, beds, baths, and living space. Show average price, median price, price per square foot, and distribution charts.

2. Compare a Listing

User enters a listing price and property details. The app compares it against similar homes and shows whether it looks low, fair, or high.

3. Predict Fair Value

Train regression models to estimate home price from housing and demographic features, then display error and confidence style ranges.

Everything else supports these pillars: filtering, charts, maps, data cleaning, and model evaluation.

Datasets Planned

The project will combine listing level and national housing trend data.

Primary Dataset: American House Prices

Kaggle dataset by Jeremy Larcher. Includes top 50 U.S. cities with price, beds, baths, living space, address, city, state, zip code, population, density, county, income, latitude, and longitude.

National Trend Dataset: FRED MSPUS

Median Sales Price of Houses Sold for the United States. Useful for showing long term national trend context from Q1 1963 to Q1 2026.

Supplemental Dataset: FRED ASPUS

Average Sales Price of Houses Sold for the United States. Useful for comparing average vs. median sales price trends.

Supplemental Dataset: House Price Index

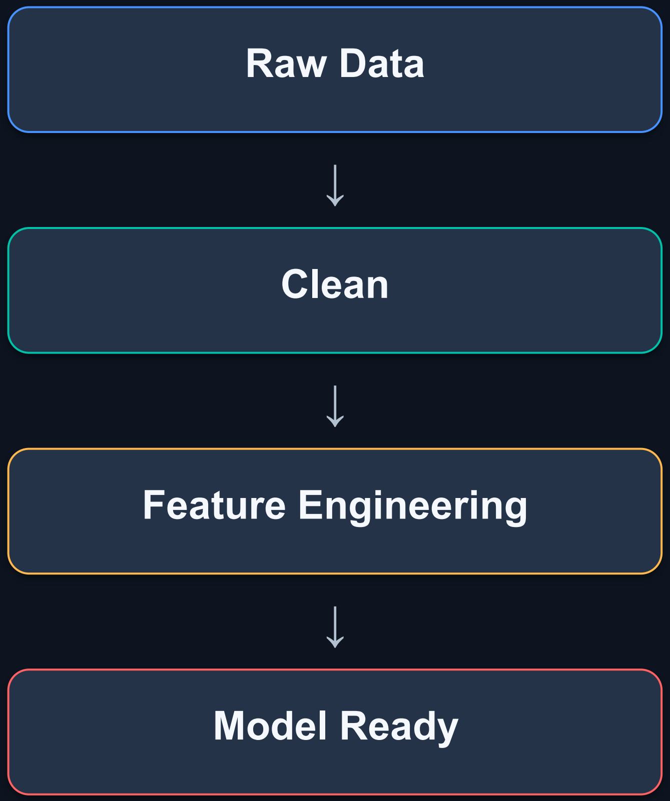
All Transactions House Price Index or FHFA style price index data can support trend charts and market context by time period.

Dataset Search sources: American House Prices, FRED MSPUS, FRED ASPUS, U.S. House Price Index datasets.

Data Preparation Plan

The technical depth starts before the model.

- Clean missing values in price, beds, baths, living space, and location fields.
- Remove or cap extreme outliers that distort the average price and model training.
- Create new features such as price per square foot, bed/bath ratio, income-to-price ratio, and density bands.
- Encode categorical fields such as state, city, county, and zip code using one-hot encoding or target encoding.
- Use train/test split and avoid data leakage when scaling or transforming features.



Machine Learning Plan

Compare multiple regression approaches instead of relying on one model.

Baseline Model

Linear Regression to establish a simple benchmark and show how much more complex models improve.

Tree-Based Model

Random Forest or Gradient Boosting Regressor to capture non-linear price patterns across location and property size.

Optional Deep Model

A small PyTorch neural network if time allows, compared against the simpler ML models.

Evaluation Metrics

- MAE: average dollar error
- RMSE: larger penalty for big misses
- R^2: overall explained variance
- Residual plots: where the model fails

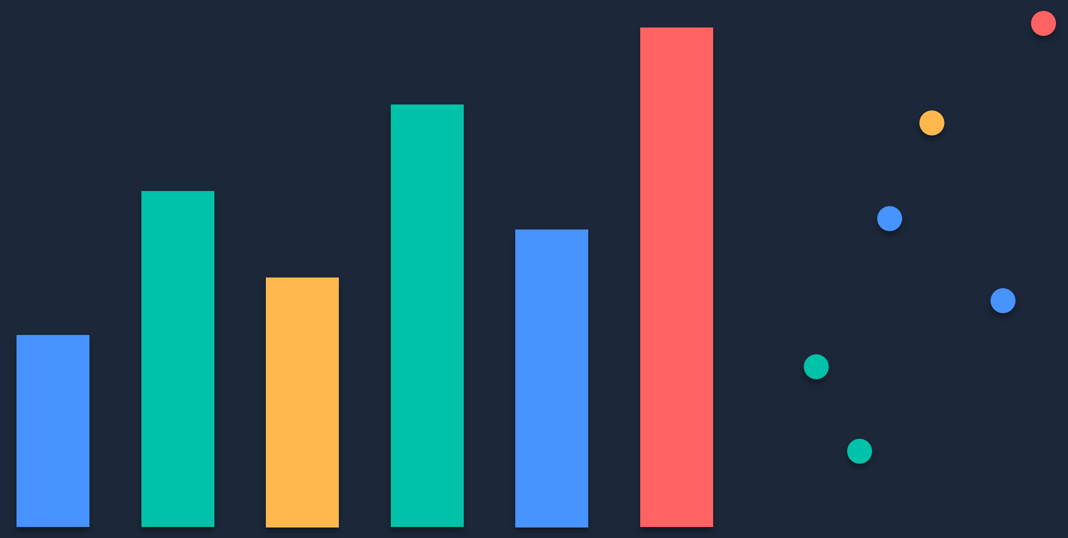
Expected proof of depth:

Model comparison table + examples where the prediction is close, too high, or too low. This proves experimentation and error analysis.

Dashboard & Visualizations

The app will make the market easier to understand visually.

Market Dashboard Mockup



Avg Price by City

Price vs. Living Space

Visuals to Include

Price distribution, price per square foot, city/state averages, trend lines, and map view using latitude/longitude.

User Facing Output

Simple labels: below average, near average, above average, or high risk overpriced.

Technical Output

Charts will support model analysis: residual plots, error by city, and feature impact summaries.

Platform, Languages & APIs

Full technology stack for the one month build.

Frontend

React + Vite or Next.js, dashboard UI, form inputs, filters, chart containers.

Backend

Python FastAPI for prediction endpoints, dataset processing, and model serving.

Data / ML

Pandas, NumPy, Scikit learn, optional PyTorch, saved model pipeline.

Visuals

Plotly, Recharts, or Chart.js for interactive charts and market comparisons.

Map / API

Leaflet/OpenStreetMap for zip/city map view. Optional Census/FRED API for trend data.

Deployment

Vercel frontend + Render/Railway backend, or Streamlit as a faster fallback.

One month fallback: Streamlit + Pandas + Scikit-learn + Plotly, if fullstack scope becomes too large.

Similar Products

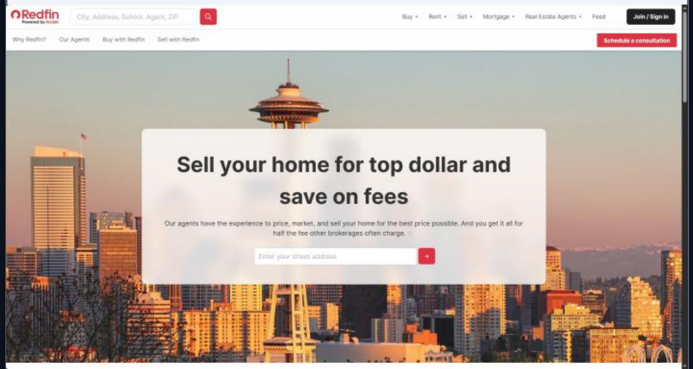
HomeScope is smaller, but more transparent about data and model results.

Zillow

Listings, market estimates, neighborhood data.

Focus: consumer search + Zestimate.

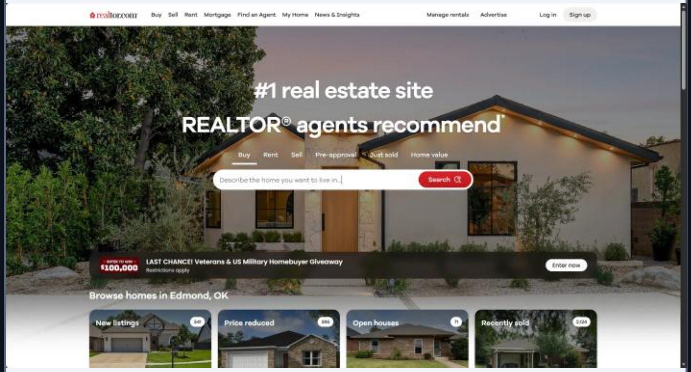
Redfin



Listings, home estimates, market dashboards.

Focus: brokerage driven search experience.

Realtor.com



Listings, property details, market resources.

Focus: MLS style property discovery.

Differentiator: HomeScope will show the actual data pipeline, model comparison, prediction error, and examples of where the model succeeds or fails.

One Month Scope

Focused build plan for Project & Portfolio IV.

Week 1

Dataset download, cleaning, EDA, feature engineering, initial charts.

Week 2

Train baseline and tree-based models; compare metrics; save best model.

Week 3

Build dashboard UI, filters, prediction form, and chart views.

Week 4

Polish, deploy, record demo, document learning outcomes and errors.

Must-have deliverables

- Working dashboard
- Average price filters
- At least two model comparisons
- Prediction form
- Model metrics and error analysis

Stretch goals

- Interactive map
- FRED trend API pull
- Feature importance view
- User saved comparisons

Learning Outcomes

What this project proves technically.

Data Visualization

Create clear charts that explain average prices, trends, price distributions, and geographic differences.

Machine Learning

Train, evaluate, and compare regression models that predict housing prices from multiple features.

Real World Data Preparation

Handle missing data, categorical fields, outliers, skewed prices, and model-ready preprocessing.

Full Stack / App Skills

Connect a dashboard UI to a backend prediction endpoint and present results in a user-friendly way.

Pitch closing:

HomeScope is a realistic, one month AI dashboard that turns housing datasets into practical market insight while demonstrating data cleaning, visualization, machine learning, and evaluation.